

Concept Bottleneck Models on Fashion-MNIST

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1 Setup

Dataset: We study Concept Bottleneck Models (CBMs) [1] on Fashion-MNIST dataset. Eight binary concepts (`is_footwear`, `has_sleeves`, `has_collar`, etc.) are assigned from the class label. Some classes share the same concept vector (sneaker and ankle boot for example), creating an inherent information bottleneck that limits pure concept-based classifiers.

Concept \ Class	0	1	2	3	4	5	6	7	8	9
is_footwear	No	No	No	No	No	Yes	No	Yes	No	Yes
is_shoe	No	No	No	No	No	No	No	Yes	No	Yes
is_sneaker	No	No	No	No	No	Yes	No	Yes	No	Yes
has_sleeves	Yes	No	No	Yes	Yes	No	No	No	No	No
has_collar	No	No	No	No	No	No	No	No	No	No
is_bag	No	No	No	No	No	No	No	No	No	No
is_hat	No	No	No	No	No	No	No	No	No	No
is_glasses	No	No	No	No	No	No	No	No	No	No
is_sunglasses	No	No	No	No	No	No	No	No	No	No

Figure 1: Concept-to-class assignment matrix. “Yes” indicates the concept is active for that class.

Architecture: Our neural network is a three-block CNN backbone (Conv–BN–ReLU–Dropout2d–MaxPool, channels 32/64/128, output 1152-d). The main objective of the project is to compare different configurations:

1. **Baseline** ($x \rightarrow y$): backbone $\rightarrow$ FC-256 $\rightarrow$ 10 classes. Does not include concept vectors.
2. **Concept predictor** ($x \rightarrow c$): backbone $\rightarrow$ FC-256 $\rightarrow$ 8 concept logits (BCE loss). Verifies learnability of the chosen concepts.
3. **Independent CBM** ($x \rightarrow c \rightarrow y$): frozen concept predictor + a small label head (FC-16 $\rightarrow$ 10).
4. **Joint CBM** ($x \rightarrow c \rightarrow y$): same topology, trained end-to-end with both classification and concept loss.
5. **Hybrid CBM** ($x \rightarrow c + x \rightarrow y$): Joint CBM plus an independent side channel that maps image features directly to class logits. Final output = $f(c) + s(x)$.

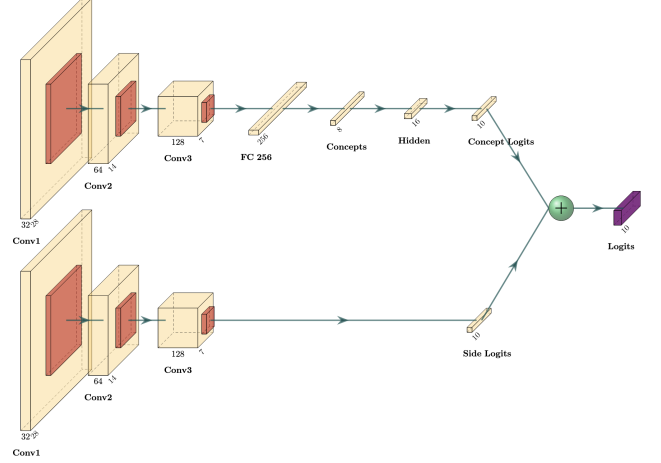


Figure 2: Hybrid CBM architecture. The concept path (top) predicts concepts and then class logits; the side channel (bottom) maps image features directly to class logits. Both are summed.

Training: We trained the model with the Adam optimiser, using a learning rate of 3×10^{-4} and a batch size of 256. We also applied random horizontal flips as data augmentation, and used early stopping with a patience of 10 epochs based on the validation loss, with an 80/20 train-validation split.

2 Results

2.1 Architecture comparison

Table 1: Test-set results for all architectures.

Model	Acc.	AUROC	Conc. F1	Ep.
Joint CBM	0.931	0.995	0.966	86
Hybrid CBM	0.928	0.996	0.964	55
Baseline	0.925	0.996	—	44
Independent CBM	0.838	0.987	0.967	37

Table 1 includes test-set performance. The Joint CBM achieves the highest accuracy, slightly

above the Baseline and Hybrid CBM. The Independent CBM is below in this metric illustrating the cost of strict interpretability. AUROC is ≥ 0.987 for all models. Concept quality is nearly identical across all CBM variants.

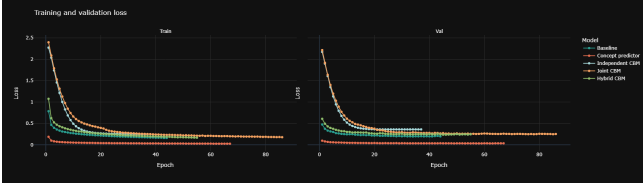


Figure 3: Training and validation loss curves for each model.

All models learn quickly in the first epochs, with the loss dropping sharply before gradually stabilising. Training and validation curves stay fairly close overall, which suggests limited overfitting. Among the models, the concept predictor reaches the lowest loss, while the others stabilize at a higher loss value.

2.2 Side-channel dropout experiment

To probe how much the Hybrid CBM relies on its direct image path, we retrain it with side-channel dropout $p \in \{0.0, 0.1, 0.3, 0.5, 0.7, 0.9\}$.

Table 2: Hybrid CBM performance vs. side-channel dropout.

Dropout	Acc.	AUROC	Conc. F1
0.0	0.927	0.996	0.965
0.1	0.928	0.996	0.963
0.3	0.928	0.996	0.964
0.5	0.929	0.996	0.965
0.7	0.929	0.996	0.966
0.9	0.932	0.996	0.967

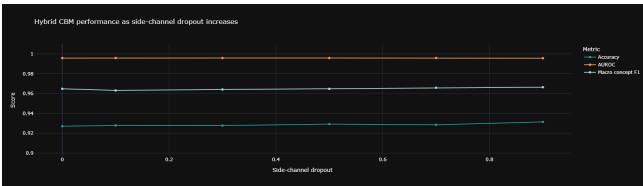


Figure 4: Hybrid CBM metrics as side-channel dropout increases. Accuracy remains stable and even peaks at $p=0.9$.

All the metrics stay nearly constant for all the dropouts, showing the model does *not* depend critically on the side channel. The concept path alone carries substantial predictive power; the side channel provides a supplement.

2.3 Concept interventions & steerability

For each CBM variant we flip one concept at a time (binarised at 0.5) and measure the average absolute change in the predicted class probability and the percentage of test samples whose label changes.

Table 3: Top-3 most influential concepts per model).

Model	Concept	Δp	Flip %
Joint	has_collar	0.756	70.5
	is_outerwear_layer	0.564	51.6
	is_legwear_or_foot.	0.476	53.7
Independent	is_outerwear_layer	0.525	70.2
	is_long_garment	0.521	50.3
	has_collar	0.454	59.9
Hybrid	is_long_garment	0.195	16.1
	has_collar	0.133	10.5
	is_closed_footwear	0.119	9.3

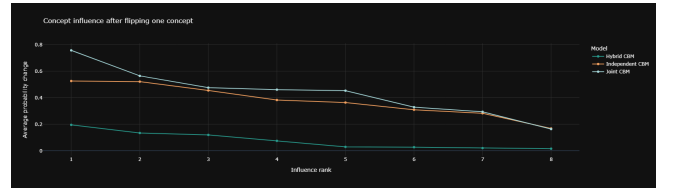


Figure 5: Average probability change vs. concept influence rank for each CBM variant. Joint CBM is the most steerable; Hybrid CBM the least.

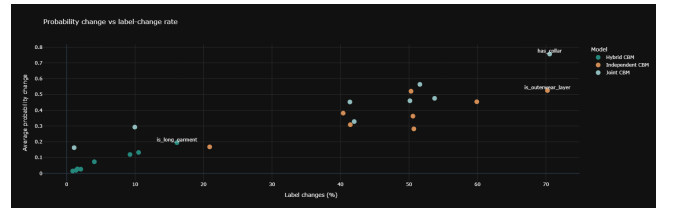


Figure 6: Probability change vs. label-flip rate per concept. The Joint and Independent CBMs cluster in the high-influence region; the Hybrid CBM is confined to the lower-left.

The **Joint CBM** is the most steerable. Changing one concept often causes a big change in prediction, so the model depends strongly on the concepts. The **Independent CBM** has medium steerability. Concept changes still affect the output a lot, but the effect is more moderate and stable. The **Hybrid CBM** is the least steerable. Changing a concept usually has a smaller effect because the model also uses a direct image path, which reduces the impact of concept interventions.

Across all models, `has_sleeves` is the least important concept, probably because it appears in many classes and does not help much to separate them. By contrast, more specific concepts like `has_collar` or `is_closed_footwear` are more influential.

Overall, concepts that produce bigger probability changes also tend to change the predicted label more often. The **Joint CBM** is the most steerable: `has_collar` alone flips the label in 70% of cases. The **Independent CBM** shows moderate steerability. The **Hybrid CBM** is the least steerable—its side channel anchors predictions, dampening concept interventions (top concept flips only 16% of labels).

3 Discussion

Accuracy vs. interpretability. The Joint CBM reaches 93.1% accuracy (above the Baseline) while keeping an inspectable concept layer. The key mechanism is that end-to-end gradients let the concept layer encode additional information, even when two classes share the same binary concept vector. This makes the concept layer less interpretable but the trade-off is strongly favourable. The Independent CBM, by contrast, pays an accuracy penalty for strict interpretability because the frozen concept predictor cannot adapt to classification needs and the label head sees only 8 binary values.

Accuracy vs. steerability. The Joint CBM offers both the best accuracy and the highest steerability, concepts tightly coupled to the classification boundary make concept flips highly impactful. The Independent CBM sacrifices ac-

curacy but retains strong steerability because the label head has no alternative information source. The Hybrid CBM recovers accuracy via the side channel but at the cost of steerability: a domain expert flipping a concept would have limited ability to correct a wrong prediction.

Concept design influence. Across all experiments, `has_sleeves` is consistently the least influential concept (rank 8 in all models). Five of ten classes activate it, making it a weak discriminator. Conversely, `has_collar` and `is_closed_footwear`—active for only two classes each—are among the most influential. This highlights that concept set design is critical: a small number of well-chosen, discriminative concepts matters more than a large set of redundant attributes.

Reference

- [1] P. W. Koh et al. Concept Bottleneck Models. Proceedings of the 37th International Conference on Machine Learning, PMLR 119, 2020.